
THE MISSING HORIZON IN RANDOM RESHUFFLING

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ABSTRACT

Random reshuffling now has an upper bound that dominates stochastic gradient descent at every stable constant step size and every finite horizon. Its matching lower bound was left open. We solve this problem for the exact all-inner average used by the upper bound. The answer is not the published horizon-independent stochastic envelope. If $\alpha = \eta nL \leq 1$, the fixed-step minimax stochastic error is

$$\Theta(\eta^2 nL \sigma_\star^2 [K^{-1} + \min\{(\alpha K)^2, 1\}]).$$

It decreases, increases, and then plateaus across three horizon regimes. Its minimum occurs at $K \asymp \alpha^{-2/3}$, before the usual mixing scale α^{-1} . For $\alpha \geq 1$, the law is $\Theta(\eta \sigma_\star^2)$. The upper bound follows from an exact finite-population bridge identity and a dimension-free control of all state memory as a Lipschitz remainder. The lower bound combines a centered common-Hessian bridge with a rectifying piecewise quadratic in two dimensions. An exact telescoping identity explains why common-Hessian quadratics alone miss the eventual floor. Together with the deterministic term, this gives a complete minimax characterization for all horizons and all stable constant steps. Optimizing the step recovers the best known tuned rate, while the fixed-step theorem reveals a nonmonotone transient hidden by that rate.

1 INTRODUCTION

Random reshuffling processes every component of a finite sum exactly once per epoch in a fresh random order. It is the common practical implementation of stochastic gradient descent, and it is often faster than sampling with replacement. A long line of work has sought to explain this advantage (Gürbüzbalaban et al., 2021; Nagaraj et al., 2019; Haochen & Sra, 2019; Mishchenko et al., 2020; Nguyen et al., 2021; Safran & Shamir, 2020; 2021).

The recent result of Liu (2026) resolves the upper-bound side for smooth convex finite sums. For any stable step $\eta \lesssim 1/L$ and any number of epochs K , it bounds the exact all-inner average by

$$\frac{D^2}{\eta nK} + \min\{\eta, \eta^2 nL\} \sigma_\star^2. \quad (1)$$

This is never slower than the corresponding bound for with-replacement SGD. The same paper identifies the matching lower characterization for all steps and horizons as an important open problem. Existing lower bounds cover small steps, sufficiently many epochs, and averages of epoch-boundary iterates (Cha et al., 2023). They do not characterize the all-inner output in (1).

A natural conjecture is that (1) is minimax-sharp for each fixed (η, K) . It is false. All-inner averaging sees a fresh without-replacement bridge in every epoch, but the epoch-end state remembers earlier bridges. Before curvature has time to rectify these fluctuations, the two effects cancel more strongly than the envelope records. The correct stochastic law for $\alpha = \eta nL \leq 1$ is

$$\eta^2 nL \sigma_\star^2 \left[\frac{1}{K} + \min\{(\alpha K)^2, 1\} \right]. \quad (2)$$

It has three phases:

horizon	dominant mechanism	stochastic error
$K \lesssim \alpha^{-2/3}$	fresh bridge	$\eta^2 n L \sigma_\star^2 / K$
$\alpha^{-2/3} \lesssim K \lesssim \alpha^{-1}$	rectification	$\eta^2 n L \sigma_\star^2 (\alpha K)^2$
$K \gtrsim \alpha^{-1}$	rectified floor	$\eta^2 n L \sigma_\star^2$

Thus more epochs can temporarily make the fixed-step minimax error worse. The minimum is smaller than the eventual floor by $\alpha^{2/3}$.

Our proof isolates the two mechanisms. For the upper bound, freeze every component gradient at a minimizer. The resulting permutation prefixes reset to zero after each epoch and have an exact bridge-area variance of $\eta^2(n+1)\sigma_\star^2/12$. Every nonlinear and state-memory effect is a single Lipschitz remainder. A prefix-sum operator controls this remainder when αK is small, and the general bound of [Liu \(2026\)](#) covers the remaining horizons.

For the lower bound, a centered common-Hessian quadratic generates the $1/K$ bridge term. It also satisfies an exact identity that makes its all-inner average proportional to its final state, so its error eventually decays rather than reaching a floor. A piecewise quadratic adapted from the one-epoch construction of [Cha et al. \(2023\)](#) rectifies the bridge into a mean displacement of order $\eta\sigma_\star\sqrt{n}\min\{\alpha K, 1\}$. Choosing the larger of these two gadgets and adding one deterministic coordinate gives a matching two-dimensional instance. Curvature rescaling handles large nominal steps without padding or inactive-component arguments.

Contributions. We give the first fixed-step minimax law for the all-inner output of random reshuffling at every finite horizon and every stable constant step. The law answers the lower-bound question in [Liu \(2026\)](#), but also shows why its upper envelope is not pointwise tight. The proof introduces a reset bridge decomposition for the all-inner transient, identifies an exact quadratic cancellation, and transfers the known rectification gadget from epoch endpoints to all-inner averaging.

2 SETUP

Let

$$f(x) = \frac{1}{n} \sum_{i=1}^n f_i(x), \quad f_i : \mathbb{R}^d \rightarrow \mathbb{R},$$

where every component is convex and L -smooth. Let $x^\star$ minimize f and define

$$\sigma_\star^2 = \frac{1}{n} \sum_{i=1}^n \|\nabla f_i(x^\star)\|^2.$$

At epoch k , random reshuffling draws an independent uniform permutation π_k and performs

$$x_{k,j+1} = x_{k,j} - \eta \nabla f_{\pi_k(j+1)}(x_{k,j}), \quad j = 0, \dots, n-1, \quad x_{k+1,0} = x_{k,n}. \quad (3)$$

We study the pre-update all-inner average

$$\bar{x}_K = \frac{1}{nK} \sum_{k=0}^{K-1} \sum_{j=0}^{n-1} x_{k,j}. \quad (4)$$

For a constant step, this is exactly the weighted output in Theorem 1 of [Liu \(2026\)](#).

For $D, \sigma \geq 0$, define the fixed-step minimax risk

$$\mathfrak{R}_{n,K}(\eta, L, D, \sigma) = \sup \mathbb{E}[f(\bar{x}_K) - f(x^\star)], \quad (5)$$

where the supremum ranges over all dimensions, all finite sums satisfying the assumptions above, and all initial points with $\|x_{0,0} - x^\star\| \leq D$ and $\sigma_\star \leq \sigma$. The lower-bound instance may depend on (η, K) , as usual in a minimax statement.

Set

$$\alpha = \eta n L, \quad \delta = \min\{\alpha, 1\}, \quad \mathcal{S}_{n,K} = \eta \sigma^2 \delta \left[\frac{1}{K} + \min\{(\delta K)^2, 1\} \right]. \quad (6)$$

For $\alpha \geq 1$, the bracket lies in $[1, 2]$, so $\mathcal{S}_{n,K} \asymp \eta \sigma^2$.

3 THE FINITE-HORIZON MINIMAX LAW

Theorem 1 (All steps and all horizons). *There are universal constants $c, C > 0$ such that for every $n \geq 8$, $K \geq 1$, $L > 0$, $D, \sigma \geq 0$, and $0 < \eta \leq 1/(6L)$,*

$$\begin{aligned} c \left[\min \left\{ LD^2, \frac{D^2}{\eta n K} \right\} + \mathcal{S}_{n,K} \right] &\leq \mathfrak{R}_{n,K}(\eta, L, D, \sigma) \\ &\leq C \left[\min \left\{ LD^2, \frac{D^2}{\eta n K} \right\} + \mathcal{S}_{n,K} \right]. \end{aligned} \quad (7)$$

The lower bound is attained in dimension at most two. For even n it uses balanced signs. For odd n the same construction uses one zero sign.

The theorem separates a deterministic optimization term from a stochastic reshuffling term. It does not assert that every instance follows the three-phase law. In fact, the exact quadratic identity in Section 5 gives an instance whose error vanishes after the bridge phase. The statement is a worst-case characterization at each fixed step and horizon.

The result also explains why the tuned rate in Liu (2026) is correct despite the slack in (1). Minimizing (7) over stable constant steps gives, up to universal constants,

$$\frac{LD^2}{nK} + \min \left\{ \frac{\sigma D}{\sqrt{nK}}, \left(\frac{L\sigma^2 D^4}{nK^2} \right)^{1/3} \right\}. \quad (8)$$

The transient dip concerns fixed steps. Tuning can move the trajectory to a different phase.

4 UPPER BOUND: A RESET BRIDGE AND ONE REMAINDER

Translate x^* to zero and write

$$g_i = \nabla f_i(0), \quad h_i(x) = \nabla f_i(x) - g_i.$$

Then $\sum_i g_i = 0$ and $\|h_i(x)\| \leq L\|x\|$. First run (3) from zero and call this trajectory z . Flatten its $N = nK$ pre-update iterates into $z_0, \dots, z_{N-1}$. Inside each epoch define the reset bridge

$$Y_{k,j} = -\eta \sum_{r=1}^j g_{\pi_k(r)}, \quad j = 0, \dots, n-1. \quad (9)$$

Completed epochs contribute zero to the frozen gradients. Therefore, after flattening $Y_{k,j}$ into Y_t ,

$$z_t = Y_t + R_t, \quad R_t = -\eta \sum_{s < t} h_{i_s}(z_s). \quad (10)$$

The bridge has exact finite-population moments.

Lemma 2 (Reset bridge). *For $j = 0, \dots, n-1$,*

$$\mathbb{E}\|Y_{k,j}\|^2 = \eta^2 \sigma_\star^2 \frac{j(n-j)}{n-1}, \quad (11)$$

$$\mathbb{E} \left\| \frac{1}{N} \sum_{t=0}^{N-1} Y_t \right\|^2 = \frac{\eta^2(n+1)}{12K} \sigma_\star^2. \quad (12)$$

Also,

$$\frac{1}{N} \sum_{t=0}^{N-1} \mathbb{E}\|Y_t\|^2 = \frac{\eta^2(n+1)}{6} \sigma_\star^2. \quad (13)$$

The $1/K$ term in (6) is exactly the variance reduction from averaging the independent centered bridge areas in (12). The second term comes from controlling R .

Lemma 3 (Short-horizon memory). *Let $\theta = \eta L N = \alpha K$. If $\theta \leq 1/2$, then*

$$\mathbb{E}\|\bar{z}_K\|^2 \leq C_0 \eta^2 n \sigma_\star^2 \left(\frac{1}{K} + \theta^2 \right) \quad (14)$$

for a numerical constant C_0 .

The proof is a one-line operator estimate. Let P be the strictly lower triangular prefix-sum matrix on N coordinates. With $a_t = \|z_t\|$ and $b_t = \|Y_t\|$, (10) gives $a \leq b + \eta L P a$. Since $\|P\|_{2 \rightarrow 2} \leq N$,

$$\|a\|_2 \leq 2\|b\|_2, \quad \|R\|_2 \leq 2\theta\|b\|_2.$$

Jensen's inequality, (12), and (13) prove (14). This argument uses no Hessians, commutativity, or dimension restriction.

Smoothness of f now gives

$$\mathbb{E}[f(\bar{z}_K) - f^\star] \leq C \eta^2 n L \sigma_\star^2 \left(\frac{1}{K} + (\alpha K)^2 \right), \quad \alpha K \leq \frac{1}{2}. \quad (15)$$

To restore an arbitrary initialization, couple x and z with the same permutations. Every component gradient map is nonexpansive when $\eta \leq 2/L$, so $\|\bar{x}_K - \bar{z}_K\| \leq D$. Co-coercivity and smoothness imply

$$f(\bar{x}_K) - f^\star \leq 2(f(\bar{z}_K) - f^\star) + LD^2. \quad (16)$$

When $\alpha K \leq 1/2$, the deterministic minimum in (7) equals LD^2 , so this proves the desired short-horizon bound.

For $\alpha K > 1/2$, Theorem 1 of Liu (2026) gives

$$\mathbb{E}[f(\bar{x}_K) - f^\star] \leq \frac{6D^2}{\eta n K} + 51\eta \sigma_\star^2 \min\{\alpha, 1\}. \quad (17)$$

If $\alpha \leq 1$, the bracket in (6) is at least $1/4$ in this region. If $\alpha \geq 1$, it is at least one. The first term in (17) is within a universal factor of the capped deterministic term. This completes the upper bound.

5 LOWER BOUND: CANCELLATION AND RECTIFICATION

The deterministic term follows from identical components $f_i(w) = \lambda w^2/2$, initialized at $w_0 = D$, with $\lambda = \min\{L, (\eta n K)^{-1}\}$. Their geometric all-inner average is a constant fraction of D , which yields

$$f(\bar{w}_K) - f^\star \geq c \min \left\{ LD^2, \frac{D^2}{\eta n K} \right\}. \quad (18)$$

It remains to obtain (6). We first expose the fresh bridge with a centered common-Hessian quadratic. Let s_i be balanced signs and set

$$Q_i(u) = \frac{\ell}{2} u^2 + a s_i u, \quad u_0 = 0. \quad (19)$$

For odd n , one sign is zero. The scale a is adjusted so that $n^{-1} \sum_i a^2 s_i^2 = \sigma^2$. Put $q = 1 - \eta \ell$ and $r = q^n$. Summing all nK update equations and using $\sum_i s_i = 0$ gives the pathwise identity

$$\boxed{\bar{u}_K = -\frac{u_{K,0}}{\eta \ell n K}.} \quad (20)$$

This exact cancellation is special to all-inner averaging.

If

$$V_n(q) = \sum_{j=1}^n \left(q^{n-j} - \frac{1}{n} \sum_{r=1}^n q^{n-r} \right)^2,$$

finite-population covariance gives

$$\mathbb{E}[Q(\bar{u}_K) - Q(0)] = \frac{\sigma^2 V_n(q)}{2\ell n(n-1)K^2} \frac{1-r^{2K}}{1-r^2} \quad (21)$$

for every n , where $Q = n^{-1} \sum_i Q_i = \ell u^2/2$. When $\beta = \eta n \ell$ and βK are small, $V_n(q) \asymp (\eta \ell)^2 n^3$ and the geometric factor is $\asymp K$. Hence

$$\mathbb{E}[Q(\bar{u}_K) - Q(0)] \geq c \frac{\eta^2 n \ell \sigma^2}{K}. \quad (22)$$

For $K \gg 1/\beta$, (21) instead decays as K^{-2} . Common-Hessian quadratics cannot create the floor.

The floor comes from curvature rectification. Consider

$$r_i(v) = \frac{1}{2} \left(\ell \mathbf{1}_{\{v < 0\}} + \frac{\ell}{2415} \mathbf{1}_{\{v \geq 0\}} \right) v^2 + a s_i v, \quad v_0 = 0. \quad (23)$$

Every component is convex and ℓ -smooth. The average objective is minimized at zero and is $\ell/2415$ -strongly convex.

Lemma 4 (Finite-horizon rectification). *There are universal constants $b, c_1, c_2 > 0$ such that, whenever $\beta = \eta n \ell \leq b$, random reshuffling on (23) satisfies, with $B = \eta a \sqrt{n}$,*

$$m_{k+1} \geq (1 - c_1 \beta) m_k + c_2 \beta B, \quad m_k = \mathbb{E}[v_{k,0}], \quad m_0 = 0. \quad (24)$$

Consequently, if $K \geq \beta^{-2/3}$, then

$$\mathbb{E}[r(\bar{v}_K) - r(0)] \geq c \eta^2 n \ell a^2 \min\{(\beta K)^2, 1\}. \quad (25)$$

The statement holds for balanced signs when n is even and for balanced signs plus one zero when n is odd.

The recurrence is the one-epoch rectification estimate proved in Appendix B of Cha et al. (2023). The new all-inner transfer is short. Couple (23) to the common-curvature ℓ quadratic under the same permutation. Monotonicity gives a pathwise lower comparison. Conditioned on an epoch start,

$$\mathbb{E}[v_{k,j} \mid v_{k,0}] \geq (1 - \eta \ell)^j v_{k,0}.$$

Solving (24) gives $m_k \geq cB \min\{\beta k, 1\}$. Averaging over the starts of the first K epochs, and using $K \geq \beta^{-2/3} > 1$, yields $\mathbb{E}\bar{v}_K \geq cB \min\{\beta K, 1\}$. Strong convexity and Jensen's inequality prove (25). Appendix D records the constants and the odd- n reduction.

To cover every nominal step, choose an effective curvature

$$\ell = \frac{\beta}{\eta n}, \quad \beta = \min\{\alpha, b\}. \quad (26)$$

Then $\ell \leq L$. Use (19) when $K \lesssim \beta^{-2/3}$ and (23) otherwise. The two bounds give

$$c \eta \sigma^2 \beta \left[\frac{1}{K} + \min\{(\beta K)^2, 1\} \right].$$

Since $\beta/\delta \geq b$ and, for $0 < \beta \leq \delta \leq 1$,

$$\frac{1}{K} + \min\{(\beta K)^2, 1\} \geq \left(\frac{\beta}{\delta} \right)^2 \left[\frac{1}{K} + \min\{(\delta K)^2, 1\} \right],$$

this is a universal constant fraction of (6). Finally, take the orthogonal direct sum of the deterministic coordinate and the selected stochastic coordinate. Block-diagonal smoothness remains at most L , the initial distance is exactly D , and the gradient variance at the minimizer is exactly σ^2 . This proves the lower bound in Theorem 1.

6 RELATION TO PRIOR RESHUFFLING THEORY

Classical analyses established improved rates for random reshuffling under small-step, large-epoch, strong-convexity, or bounded-gradient conditions (Gürbüzbalaban et al., 2021; Nagaraj et al., 2019; Haochen & Sra, 2019; Mishchenko et al., 2020; Nguyen et al., 2021). Lower bounds for final iterates and quadratic problems reveal condition-number and epoch transitions (Safran & Shamir, 2020; 2021). Last-iterate and primal-dual analyses have since sharpened the output guarantees (Liu & Zhou, 2024; Cai et al., 2024).

The closest lower bound is Cha et al. (2023). It treats arbitrary nonnegative combinations of epoch-boundary iterates. Its convex corollary uses a small step and a sufficiently large horizon. We use its one-epoch piecewise-quadratic estimate, but our output contains every inner iterate and our result tracks every finite horizon. The bridge term and the transient dip have no analogue in an endpoint-only average.

The closest upper bound is Liu (2026). It introduced the exact all-inner output in (4), proved (17) at all stable steps and horizons, and explicitly left the corresponding complete lower characterization open. Our theorem closes that direction. It also shows that the stochastic part of (17) is loose by an unbounded factor at $K \asymp \alpha^{-2/3}$ as $\alpha \rightarrow 0$.

Constant-step stationary-distribution and bias expansions study a different asymptotic object, usually epoch endpoints under strong monotonicity or additional structure (Emmanouilidis et al., 2026). They are complementary to the finite-horizon convex minimax law here.

7 DISCUSSION

The nonmonotonicity is an output effect, not a claim that running a fixed instance longer must hurt. Minimax instances can depend on the horizon. The law says which mechanism an adversary can activate at each horizon. It also clarifies the role of state memory. Memory creates no fourth minimax scale. During the transient it is contained by the Lipschitz remainder, while in the common-Hessian lower gadget it cancels the bridge exactly.

Our theorem uses the maximum component smoothness L . A sharper law that separates average and maximum smoothness, as in the nonuniform upper theorem of Liu (2026), remains open. The result also concerns the specific all-inner average. Other output rules can have different horizon laws.

REPRODUCIBILITY STATEMENT

All assumptions and complete proofs are included in the manuscript. The supplementary verification script checks the exact quadratic identity by permutation enumeration and Monte Carlo simulation, including the pre-update indexing convention. No external data or specialized hardware is required.

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A FINITE-POPULATION IDENTITIES

We prove Lemma 2. Let $G_1, \dots, G_n \in \mathbb{R}^d$ satisfy $\sum_i G_i = 0$ and $n^{-1} \sum_i \|G_i\|^2 = \sigma_G^2$. If π is uniform, then

$$\mathbb{E}\langle G_{\pi(r)}, G_{\pi(s)} \rangle = \begin{cases} \sigma_G^2, & r = s, \\ -\sigma_G^2/(n-1), & r \neq s. \end{cases}$$

It follows directly that

$$\mathbb{E} \left\| \sum_{r=1}^j G_{\pi(r)} \right\|^2 = \sigma_G^2 \frac{j(n-j)}{n-1}. \quad (27)$$

Taking $G_i = g_i$ proves (11). Summing (27) over $j = 0, \dots, n-1$ and using

$$\sum_{j=0}^{n-1} j(n-j) = \frac{n(n-1)(n+1)}{6}$$

proves (13).

For the area of one epoch,

$$A_\pi = \frac{1}{n} \sum_{j=0}^{n-1} Y_j = -\frac{\eta}{n} \sum_{r=1}^{n-1} (n-r) g_{\pi(r)}.$$

For arbitrary scalar weights w_r , the covariance above gives

$$\mathbb{E} \left\| \sum_{r=1}^n w_r g_{\pi(r)} \right\|^2 = \frac{n\sigma_\star^2}{n-1} \sum_{r=1}^n (w_r - \bar{w})^2. \quad (28)$$

With $w_r = n - r$ and $\sum_r (w_r - \bar{w})^2 = n(n^2 - 1)/12$, we obtain

$$\mathbb{E}\|A_\pi\|^2 = \frac{\eta^2(n+1)}{12}\sigma_\star^2.$$

The epoch areas are independent and centered. Their average has variance $1/K$ times this quantity, proving (12).

B COMPLETE UPPER-BOUND PROOF

We first prove Lemma 3. Index the $N = nK$ pre-update iterates from zero. Let $P \in \mathbb{R}^{N \times N}$ have entries $P_{ts} = 1$ if $s < t$ and zero otherwise. Equation (10) and $\|h_i(z)\| \leq L\|z\|$ imply componentwise

$$a \leq b + \eta L P a, \quad a_t = \|z_t\|, \quad b_t = \|Y_t\|.$$

Since $\|P\|_{2 \rightarrow 2} \leq \|P\|_F < N$ and $\theta = \eta L N \leq 1/2$,

$$\|a\|_2 \leq \|b\|_2 + \theta \|a\|_2 \leq 2\|b\|_2. \quad (29)$$

Likewise, the sequence of remainder norms satisfies

$$\|R\|_2 \leq \eta L \|P a\|_2 \leq 2\theta \|b\|_2. \quad (30)$$

Jensen's inequality gives

$$\mathbb{E}\|\bar{R}\|^2 \leq \frac{1}{N} \mathbb{E}\|R\|_2^2 \leq 4\theta^2 \frac{1}{N} \mathbb{E}\|b\|_2^2 = \frac{2(n+1)}{3} \eta^2 \theta^2 \sigma_\star^2.$$

Together with (12) and $\|u+v\|^2 \leq 2\|u\|^2 + 2\|v\|^2$, this proves (14) with, for example, $C_0 = 4$ for $n \geq 2$.

For completeness, we justify the initialization coupling. If $T_i(x) = x - \eta \nabla f_i(x)$, co-coercivity gives

$$\|T_i(x) - T_i(y)\|^2 \leq \|x - y\|^2 - (2\eta/L - \eta^2) \|\nabla f_i(x) - \nabla f_i(y)\|^2.$$

Thus T_i is nonexpansive for $\eta \leq 2/L$. Coupled trajectories remain within D , and so do their averages. For any u, v , smoothness gives

$$f(u) - f^\star \leq f(v) - f^\star + \langle \nabla f(v), u - v \rangle + \frac{L}{2} \|u - v\|^2.$$

The standard inequality $\|\nabla f(v)\|^2 \leq 2L(f(v) - f^\star)$ and Young's inequality yield (16).

If $\theta \leq 1/2$, then $D^2/(\eta n K) = LD^2/\theta \geq 2LD^2$, so the capped deterministic term is LD^2 . Equations (15) and (16) give the upper half of (7) in this region.

If $\theta > 1/2$, apply (17). When $\theta \in (1/2, 1)$, $D^2/(\eta n K) < 2LD^2$, and when $\theta \geq 1$ it is the smaller deterministic term. Hence the first term in (17) is at most twelve times the capped term. If $\alpha \leq 1$, then

$$K^{-1} + \min\{(\alpha K)^2, 1\} \geq 1/4.$$

If $\alpha \geq 1$, then $\delta = 1$ and the same bracket is at least one. This absorbs the stochastic term in (17) and completes the proof.

C EXACT COMMON-HESSIAN CANCELLATION

We derive (21) and its lower consequence. First assume n is even and $s_i \in \{-1, 1\}$ are balanced. For one epoch,

$$u_{k+1,0} = r u_{k,0} + \xi_k, \quad \xi_k = -\eta a \sum_{j=1}^n q^{n-j} s_{\pi_k(j)}.$$

The innovations are independent and centered. Equation (28) gives

$$\mathbb{E}\xi_k^2 = \frac{\eta^2 n a^2}{n-1} V_n(q).$$

Starting from zero,

$$\mathbb{E}u_{K,0}^2 = \frac{\eta^2 na^2}{n-1} V_n(q) \sum_{k=0}^{K-1} r^{2k}.$$

The sum of the nK scalar update equations is

$$u_{K,0} - u_{0,0} = -\eta\ell \sum_{k,j} u_{k,j} - \eta a \sum_{k,j} s_{\pi_k(j)}.$$

The last term vanishes epoch by epoch. This proves (20). Substitution into $Q(\bar{u}_K) - Q(0) = \ell \bar{u}_K^2/2$ proves (21).

We record explicit bounds. If $\beta = \eta n\ell \leq 1/4$, then $q^{n-1} \geq 1 - \beta \geq 3/4$. The pairwise variance identity gives

$$\begin{aligned} V_n(q) &= \frac{1}{2n} \sum_{p,r=0}^{n-1} (q^p - q^r)^2 \\ &\geq \frac{q^{2n-2}(\eta\ell)^2}{2n} \sum_{p,r=0}^{n-1} (p-r)^2 \geq c(\eta\ell)^2 n^3. \end{aligned}$$

If also $\beta K \leq 1/4$, then $r^{2k} \geq (1-\beta)^{2K} \geq c$ for $k < K$, so $\sum_{k < K} r^{2k} \geq cK$. Equation (21) now proves (22).

For odd n , choose $(n-1)/2$ signs of each type and one zero. Put $\rho_n = n^{-1} \sum_i s_i^2 = (n-1)/n$ and $a = \sigma/\sqrt{\rho_n}$. The weighted covariance formula has $a^2 \rho_n = \sigma^2$ in place of a^2 , so the same calculation and constants apply.

D THE RECTIFICATION LEMMA

We give the promised reduction to the proved one-epoch estimates of Cha et al. (2023). Their Appendix B considers exactly (23) for even n , with a denoted by ν . Lemmas B.2–B.4 and the calculation following them show, for $\beta \leq 1/161$, that an epoch starting from a fixed $v \geq 0$ satisfies

$$\mathbb{E}[v^+ | v] \geq (1 - 3\beta/4)v + \frac{\eta^2 \ell a n^{3/2}}{18000}, \quad (31)$$

while a negative start satisfies

$$\mathbb{E}[v^+ | v] \geq (1 - 160\beta/161)v. \quad (32)$$

They also prove that, from a nonnegative initialization, every epoch start is nonnegative with probability at least $1/2$. Since $v < 0$ reverses the comparison between the two contraction coefficients, conditioning on the sign of the start gives

$$m_{k+1} \geq (1 - 3\beta/4)m_k + \frac{\beta B}{36000}, \quad B = \eta a \sqrt{n}. \quad (33)$$

This is (24). Taking a smaller universal b if needed, iteration from $m_0 = 0$ yields

$$m_k \geq cB \min\{\beta k, 1\}. \quad (34)$$

We next transfer endpoint drift to the all-inner output. Let $\tilde{v}_j$ be the trajectory with the same linear signs and common curvature ℓ . For $\eta\ell \leq 1$, both component update maps are nondecreasing and

$$T_i^{\text{rect}}(x) \geq T_i^{\text{quad}}(x) \quad \text{for every } x.$$

Induction gives $v_j \geq \tilde{v}_j$ pathwise. Conditional on the epoch start, uniform permutation marginals and balanced signs give

$$\mathbb{E}[\tilde{v}_j | v_0] = (1 - \eta\ell)^j v_0.$$

Therefore

$$\mathbb{E}v_{k,j} \geq (1 - \eta\ell)^j m_k \geq (1 - \beta)m_k.$$

If $K \geq \beta^{-2/3}$ and b is small, then $K \geq 2$. Averaging (34) over $k = 0, \dots, K-1$ gives

$$\mathbb{E}\bar{v}_K \geq cB \min\{\beta K, 1\}. \quad (35)$$

The average objective in (23) is $\ell/2415$ -strongly convex and minimized at zero. Jensen's inequality gives

$$\mathbb{E}[r(\bar{v}_K) - r(0)] \geq \frac{\ell}{4830} (\mathbb{E}\bar{v}_K)^2,$$

which proves (25).

For odd n , use balanced $+1$ and -1 signs and one zero. We explain why the cited one-epoch proof still applies. If S_j is a prefix sum of a uniform permutation of these signs, sampling without replacement gives, for $j \leq n/2$,

$$\mathbb{E}S_j^2 \geq cj, \quad \mathbb{E}S_j^4 \leq Cj^2.$$

Hölder's inequality and Paley–Zygmund imply $\mathbb{E}|S_j| \geq c\sqrt{j}$ and $\Pr(S_j > 0) = \Pr(S_j < 0) \geq c$. These are the only sign-distribution inputs in Lemmas B.2–B.4 of Cha et al. (2023). Repeating their displayed inequalities with the new universal constants proves (31)–(32) up to constant factors. Scaling a by $\sqrt{n/(n-1)}$ preserves $n^{-1} \sum_i a^2 s_i^2 = \sigma^2$ exactly.

E COMPLETING THE LOWER BOUND

Choose a universal $b > 0$ small enough that $b \leq 1/161$ and $b^{1/3} \leq 1/4$. Let $\beta = \min\{\alpha, b\}$ and choose ℓ as in (26). Then $\ell \leq L$ and $\eta n \ell = \beta$.

If $K \leq \beta^{-2/3}$, then $\beta K \leq \beta^{1/3} \leq 1/4$ and Appendix C gives

$$\mathbb{E}[Q(\bar{u}_K) - Q(0)] \geq c\eta^2 n \ell \sigma^2 / K.$$

In this regime, $(\beta K)^2 \leq 1/K$, so this is a constant fraction of

$$\eta^2 n \ell \sigma^2 [K^{-1} + \min\{(\beta K)^2, 1\}].$$

If $K \geq \beta^{-2/3}$, Lemma 4 applies and $K^{-1} \leq (\beta K)^2$. Its lower bound is again a constant fraction of the same display. Thus one of the two one-dimensional stochastic gadgets always supplies

$$c\eta\sigma^2\beta [K^{-1} + \min\{(\beta K)^2, 1\}]. \quad (36)$$

Let $\delta = \min\{\alpha, 1\}$. We have $\beta/\delta \geq b$. For $r = \beta/\delta \leq 1$,

$$\min\{(\beta K)^2, 1\} \geq r^2 \min\{(\delta K)^2, 1\}, \quad K^{-1} \geq r^2 K^{-1}.$$

Hence (36) is at least $cb^3 \mathcal{S}_{n,K}$.

Finally, verify (18). Put $T = nK$ and $\lambda = \min\{L, (\eta T)^{-1}\}$. The pre-update iterates are $w_t = (1 - \eta\lambda)^t D$, and

$$\bar{w}_K = D \frac{1 - (1 - \eta\lambda)^T}{T\eta\lambda}.$$

If $\eta LT \leq 1$, then $\lambda = L$ and $\bar{w}_K \geq cD$. If $\eta LT \geq 1$, then $\lambda = (\eta T)^{-1}$ and again $\bar{w}_K \geq cD$. Multiplying by $\lambda/2$ proves (18). The orthogonal direct sum described in the main text finishes the proof of Theorem 1.

F OPTIMIZING THE STEP

We briefly derive (8). A stable step of order $1/L$ contributes the baseline $LD^2/(nK)$. In the regime $\alpha \geq 1$, balancing $D^2/(\eta n K)$ with $\eta\sigma^2$ gives $\sigma D/\sqrt{nK}$. In the small-step regime after rectification has saturated, balancing $D^2/(\eta n K)$ with $\eta^2 n L \sigma^2$ gives $(L\sigma^2 D^4/(nK^2))^{1/3}$. Taking the better admissible balance and the stability boundary yields (8). The lower half of Theorem 1 shows that the same expression is minimax-sharp for constant-step random reshuffling.